Adding Two-Digit Numbers with Place Value

Answer Key

Grade 2–3 Arithmetic

Quick Answers

1. 38	3. 65	5. 63	7. 75
2. 59	4. 84	6. 90	8. 73

Curriculum

Level: Grade 2–3 Arithmetic
2.NBT.B.5 / 3.NBT.A.2 – *problems 1, 2, 3, 4, 5, 6, 7, 8*
Difficulty 1–4 of 5 across 8 tagged checks.

Common wrong answers

6. *If they got 92: added the exact numbers instead of rounding each one first*
6. *If they got 80: rounded 58 down to the ten below instead of to the nearest ten*
8. *If they got 613: added the tens and the ones as if they were separate little sums and wrote the two results side by side*

How to use this key. Each solution is written the way a second or third grader should say it out loud: name the tens, name the ones, put them back together. If a child gets the right total by counting on by ones, the answer is correct but the strategy is not yet there — ask them to show it again with the blocks or the number line.

1. Base-ten blocks: a group of 23 (two tens, three ones) plus a group of 15 (one ten, five ones).
Solution: Count the tall bars first: two tens and one ten make three tens, which is 30. Now the small squares: three ones and five ones make eight ones. Three tens and eight ones is 38. Written as a number sentence, $23 + 15 = 38$. The total is $\boxed{38}$.

2. Number line: start at 34 and add 25.
Solution: Break 25 into its tens and its ones: 25 is two tens and five ones. Jump the tens first — from 34 to 44, then from 44 to 54. Now jump the five ones: 54 to 59. Landing on 59 means $34 + 25 = 59$, so the total is $\boxed{59}$. Jumping the tens first is what keeps the count short: two big jumps and one small one, instead of twenty-five little steps.

3. Base-ten blocks with a trade: 38 (three tens, eight ones) plus 27 (two tens, seven ones).
Solution: Tens first: three tens and two tens make five tens, or 50. Ones next: eight ones and seven ones make fifteen ones. Fifteen ones is more than a full ten, so circle ten of them and trade: that is one more ten and five ones left over. Now there are six tens and five ones. Putting it together, $50 + 15 = 65$, so $38 + 27 = \boxed{65}$.

4. Number line: start at 46 and add 38.
Solution: 38 is three tens and eight ones. Jump the three tens: 46 to 56, 56 to 66, 66 to 76. Then jump eight ones. Going from 76 to 84 crosses a ten, so it helps to split the eight: four ones gets to 80, and four more gets to 84. The landing number is the total: $46 + 38 = \boxed{84}$.

5. Friendly ten: Rosa adds 39 and 24 by sliding 1 across.
Solution: 39 is only one away from 40, and 40 is easy to add to. So Rosa moves 1 out of the 24: the 39 becomes 40 and the 24 becomes 23. Nothing was created or lost — the same total is just packed differently. Now the addition is friendly: $40 + 23 = 63$. Checking the other way, $39 + 24 = 63$ as well, so the total is $\boxed{63}$. Note that Rosa does not have to give anything back at the end: she moved the 1 from one addend to the other rather than adding it on top.

6. Estimate: round 58 and 34 to the nearest ten, then add.
Solution: On the number line 58 sits past the middle of the way from 50 to 60, so it is closer to 60 — round it up. For 34, the 4 ones are less than half of a ten, so 34 rounds down to 30. Adding the rounded numbers, $60 + 30 = 90$, so the sum is about $\boxed{90}$. The exact answer is 92, which is close to the estimate — that closeness is how an estimate checks an answer.

7. Friendly ten on the second number: 47 plus 28.
Solution: 28 is two away from 30, so give it 2 and add the easier problem: $47 + 30 = 77$. But this time the 2 was borrowed from nowhere — it was added on top of the real total, so it has to be taken back off: $77 - 2 = 75$. Hence $47 + 28 = \boxed{75}$. Compare this with problem 5: there the 1 was moved *between* the two numbers, so nothing was given back; here the 2 was added *in*, so it must be removed.

8. Find the mistake: a student wrote $46 + 27 = 613$.
Solution: The student added the tens as if they were single digits $(4 + 2 = 6)$ and the ones separately $(6 + 7 = 13)$, then wrote the two results next to each other as "613". The 6 was really six tens, that is 600 in the way it was written, and the two pieces were never actually added together. Doing it with place value instead: six tens and seven ones is thirteen ones, so trade ten of them for a ten. That gives $40 + 20 + 10 = 70$ and three ones left over. So $46 + 27 = \boxed{73}$. A quick estimate would have caught it: 46 and 27 are close to 50 and 30, so the answer has to be near 80, nowhere near 613.